Lab Experiment-21

Aim

To write a Python program to perform syntax-driven semantic analysis by extracting noun phrases and identifying their meanings from a given sentence using spaCy.

Algorithm

1. Import the spaCy library.
2. Load the English language model.
3. Get a sentence from the user.
4. Process the sentence using spaCy.
5. Extract noun phrases using `doc.noun_chunks`.
6. Display the noun phrases and their meanings using WordNet.
7. Display the result.

Python Code

```
import spacy

from nltk.corpus import wordnet

import nltk

nltk.download('wordnet')

nlp = spacy.load("en_core_web_sm")

sentence = input("Enter a sentence: ")

doc = nlp(sentence)

print("\nNoun Phrases and Meanings:")

for chunk in doc.noun_chunks:
```

```
    print("\nNoun Phrase:", chunk.text)

    meanings = wordnet.synsets(chunk.root.text)

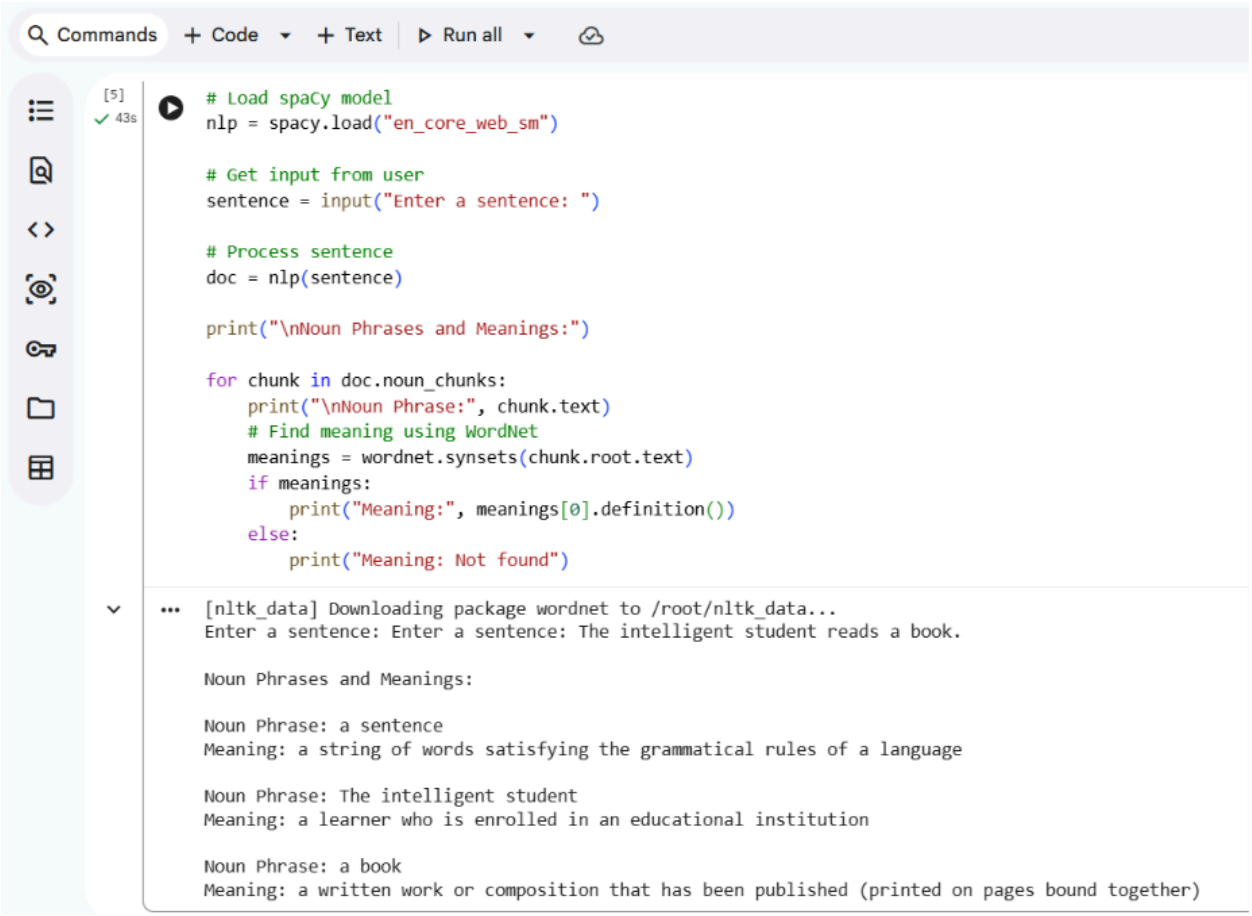
    if meanings:

        print("Meaning:", meanings[0].definition())

    else:

        print("Meaning: Not found")
```

Output:



Result

Thus, the program successfully extracts noun phrases and retrieves their meanings.